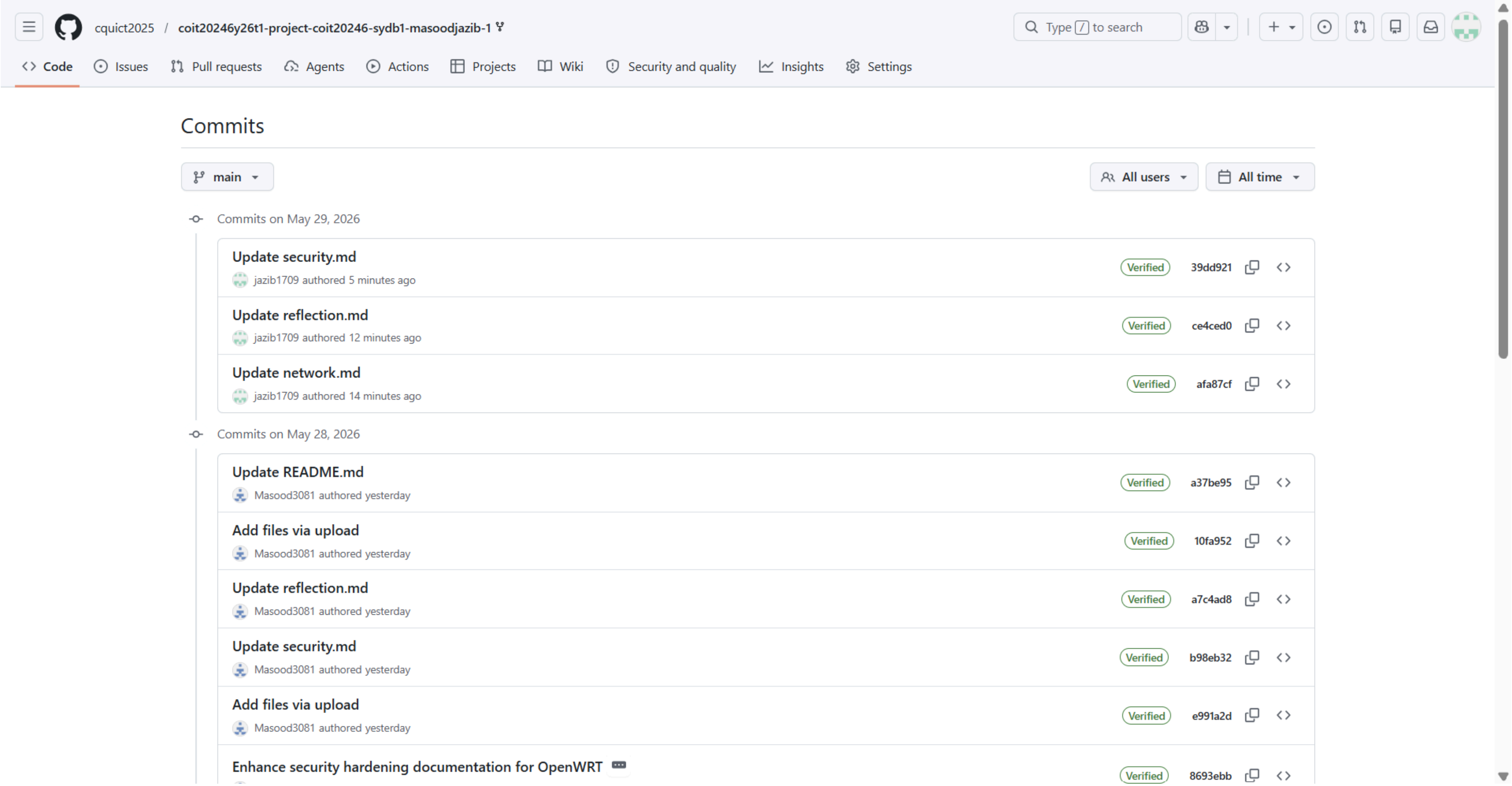
Project Reflection

GitHub Commits

The record of contributions to the project and the group was presented in the form of the GitHub commit history. As the project work proceeded, it was updated with the Markdown files, screenshots, diagrams, packet captures, and risk assessment files.



List of Tasks

Student	Student ID	Main Contributions
Jazib Hussain	12311998	OpenWRT setup, VirtualBox network documentation, website setup, firewall testing, network screenshots, lab network diagram, production network diagram, and `network.md` documentation

Student	Student ID	Main Contributions
Syed		
Masood Ali Syed	12316372	OpenWRT hardening, SSH key-based authentication, service hardening, HTTP and SSH traffic captures, Wireshark analysis, risk assessment, security controls, and `harden.md` / `security.md` documentation

Task	Main Contributor	Evidence
Project planning	Both students	`plan.md`
Business assumptions	Both students	`network.md`
OpenWRT interface documentation	Jazib Hussain Syed	`openwrt-ip-addr.png`
VirtualBox adapter documentation	Jazib Hussain Syed	`virtualbox-adapter-1.png`, `virtualbox-adapter-2.png`
Website setup	Jazib Hussain Syed	`website-browser-success.png`
Firewall configuration	Jazib Hussain Syed	HTTP, ICMP, SSH, and management interface screenshots
Lab network diagram	Jazib Hussain Syed	`lab-network.drawio`, `lab-network-diagram.png`
Production network design	Both students	`production-network.drawio`, `production-network-diagram.png`
Root password and password hash review	Masood Ali Syed	`root-password-change.png`, `etc-shadow-hash.png`
SSH key-based authentication	Masood Ali Syed	`ssh-key-generation.png`, `ssh-key-login-success.png`
Disable unnecessary service	Masood Ali Syed	`enabled-services-before.png`, `service-disabled-after.png`
HTTP traffic capture	Masood Ali Syed	`http-capture.pcap`, Wireshark HTTP screenshots
SSH traffic capture	Masood Ali Syed	`ssh-capture.pcap`, Wireshark SSH screenshots

Task	Main Contributor	Evidence
Risk assessment	Both students	`risk-assessment.xlsx`
Security controls	Both students	`security.md`
Project reflection	Both students	`reflection.md`

Reflection on Commits and Tasks

The commits do not necessarily always give the exact number of works done by respective students. There were some tasks that involved practical testing and screenshots and screenshots and troubleshoots, and file uploads and some tasks had fewer commits but still had important work. As an example, testing of firewalls, SSH setup and traffic capture involved a number of technical procedures before the eventual evidence could be integrated into the repository.

The completed tasks are associated with the commits as the repository was revised as the real-life was documented and the practicals were prepared. This project had Markdown report files, screenshots, draw.io diagrams, packet capture files and the risk assessment spreadsheet. These files demonstrate the viable practice done by the group.

The team collaborated throughout the life of the project, with initial preparation of the plan and repository, network configuration, firewall setup, hardening, and traffic analysis, risk evaluation, and final reflection. The record of contributions would have been better with more frequent contributions by both students. To improve the reflection of individual progress in the GitHub history, in future work, the group will need to commit themselves after every task accomplished.

Reflection on Group Work

The team operated with breaking up the project into practical activities and documentation activities. The real work was done prior to the written report in order to be able to base the report upon real screenshots, real IP addresses, rules tested on the firewall, and traffic captured. This contributed towards an improved accuracy and consistency of the report with the lab scenario.

A problem that was faced was locating the right active directory of the activity. It was serving the site out of /srv/www/index.html, as opposed to what the directory of the site was meant to serve, which was /www/index.html. This made the old page show up till the right web directory was found and changed.

Problem of SSH set up was also another problem. Switching of SSH ports and implementation of key-based authentication had to be closely monitored since key port settings led to errors of connection refused. This was resolved by looking at the Dropbear settings, re-initiating the service and attempting SSH access once again.

The issue of traffic capture also needed attention since, in the HTTP capture, there were 304 Not Modified responses that were in the cache. This demonstrated still the metadata of the HTTP request and response but demonstrated as well that browser caching may interfere with packet-analysis.

In future group projects, the group must be equipped with a common checklist initially, and keep it updated upon completion of each practical activity. Technical problems should also be documented as they occur with the proper paths of files, IP addresses and ports, and command. This would minimize the number of times they have to do troubleshooting, and the final report would be easier to prepare.

On the whole, this project helped the group to better understand openWRT networking, the design of the virtualbox lab, firewall testing, SSH hardening, capturing of traffic, the analysis of Wireshark, and the significance of aligning technical evidence to written documentation.